

Project Contract

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5.Project Title: Phishing Email Detection

6.Project Proposer: Dunja Majstorovic

7.Supervisor: Dunja Majstorovic

8. Introduction:

Phishing is one of the most actively employed approaches to cyberattacks through e-mails, which is based on misleading messages and social engineering tricks to trick the receiver. This project will explore the existing phishing techniques such as spear phishing, clone phishing, and whaling, and investigate the signs that are presented in these attacks. It also analyses current defense techniques, including filtering techniques to data-driven detection techniques. This project will focus on creating a program that will be able to process e-mail files and extract features that are related to specific phishing attacks and help create a more comprehensive picture of how this menace may be detected and countered.

9.Project Background:

E-mail is one of the highly used communication channels in organizations and thus a common target by threat actors. Phishing attacks have been on the increase because they can mimic the trusted sources, control the behavior of the users, and overcome conventional protection mechanisms. Phishing e-mails have become a significant problem in businesses today because they are involved in theft of credentials, unauthorised access, loss of data and even money. This has led to the necessity of better detection systems that can study incoming e-mails better and assist in the current security measures.

Old fashioned spam filters and signature based applications are used to identify the known threats and many times fail to identify the new phishing attacks, particularly the ones that are personalized, domain spoofing or the well-constructed fraudulent sites. The methods of attackers are constantly developed, and they are based on linguistic manipulation, misleading URLs and malicious attachments that look like ordinary files. Due to this changing threat environment, organizations need solutions not only that consider surface indications, but also learn about patterns based on actual phishing e-mails.

The project is the result of a functional cybersecurity requirement: to perform an early detection of phishing attacks prior to their reaching their end users. Automated e-mail filtering is common to many companies, however, several companies are making benefits by using better models which includes machine learning, feature analysis and attacker behavior insight. The project will help ensure more effective e-mail evaluation and decision-making in

security operations by analyzing sender information, URL format, contents of the messages, the type of attachments, and the details of the headers.

Moreover, the project is a response to the growing requirement in the tools to be able to conduct the review of e-mails in different formats and identify the signs of possible threats. The project aims at benefiting the operational environment by providing safer communication conditions and reinforced resistance to phishing attacks through the research of phishing strategy and defenses, as well as the creation of a detection mechanism.

10.Aims:

This project focuses on researching the existing phishing methods and determining the signs, which are evidenced in fraudulent e-mails. The objective of the project is also to evaluate the existing defence strategies, including both the conventional filtering schemes and machine learning-driven detection systems. On this understanding, the project will create a tool that can analyse e-mail files and point out features that can be used to identify the presence of phishing activities. It is aimed at aiding the more educated identification by analyzing aspects of the URLs, attachments, sender details, and message content.

11.Objectives:

The project will provide a working system of analyzing and detecting phishing emails, which will be backed up by the methods of data analysis and cybersecurity assessment. The quantifiable targets include:

1. Requirements Analysis: Determine system requirements in phishing detection, such as allowed email formats, feature extraction requirements and performance expectations.
2. Literature Research: Survey the latest phishing trends, social engineering techniques and detection techniques to guide the system design and functionality choice.
3. Data Collection: Obtain marked sets of phishing and legitimate email messages in publicly accessible sources to train the model and test it.
4. Feature Engineering: Engineer features of the emails, including sender, URLs, attachments, headers and textual messages that are indicative of phishing.
5. Model Development: Deploy machine learning or text-analysis models that can categorize emails as phishing or legitimate depending on features detected in the email.
6. Cybersecurity Analysis: Discuss the tactics of attackers and social engineering to improve the detection threshold and relevance of systems.
7. Tool Development: Develop a software that has the ability to take email files, execute the detection model, and give results that identify possible phishing red flags.
8. Testing and Evaluation: Check the functionality of the system, such as the accuracy of detection, false positive test results, and consistency under a variety of phishing conditions.

9. Documentation and Recommendations: Prepare a detailed documentation, such as the methodology, results, and recommendations to improve the system and continue working on it.

12.Deliverables:

The following are the deliverables that will be generated by the project:

1. Requirements Specification: Document with specifications of system requirements, allowed email format, and feature extraction specifications.
2. Literature Review Report: Overview of phishing activities, social engineering strategies, and detection methods.
3. Email Dataset: Phishing and legitimate emails were collected and labelled to be analyzed.
4. Feature Analysis Report: Features identified and extracted the most important email features that could be used to detect phishing.
5. Detection Tool: Software that is able to scan emails, use a detection model, and point out possible phishing indicators.
6. Testing and Evaluation Report: The tool performance according to the accuracy and reliability measures.
7. Final Documentation: Complete project report such as methodology, source code, results and recommendations.

13. Resources and Constraints:

Resources:

- Being able to train a model requires adequate processing power on a personal computer.
- Python programming environment (Python, scikit-learn, pandas, NLTK, or TensorFlow).
- Publicly available phishing and authentic email data.

Constraints:

- Scarcity of big and current labeled collections of emails.
- Limitations of time in terms of testing and evaluation.
- Any possible differences in email formats that can influence feature extraction.
- Net Reliance on software library compatibility and hardware capacity to process efficiently.

14. Sources of Information:

- IEEE Xplore, ACM Digital Library, SpringerLink on phishing and email security
- *Cybersecurity for Beginners* by Raef Meeuwisse; *Machine Learning for Cybersecurity* by Kumar et al.
- Kaggle phishing datasets, PhishTank, VirusTotal, cybersecurity blogs

- ENISA, NIST, Anti-Phishing Working Group (APWG)
- Python libraries (scikit-learn, TensorFlow, NLTK), email protocols (SMTP, MIME)

15. Risk Analysis:

- **Limited Dataset Availability:** If sufficient phishing and legitimate email samples are not available, model training may be impacted.
mitigation: combine multiple public datasets and generate synthetic samples.
- **Hardware/Software Limitations:** Large datasets may exceed processing capacity;
mitigation: use cloud resources or optimize data processing.
- **Model Performance Issues:** Low accuracy or high false positives may occur;
mitigation: iteratively refine features, adjust algorithms, and perform cross-validation.
- **Tool Compatibility:** Software may not handle all email formats.
mitigation: include support for common formats and document limitations.

16. Schedule of Activities:

The project will follow a phased schedule that aligns with the standard progression of research and development work. During the initial phase, time will be dedicated to confirming the project scope, aligning expectations with the supervisor, and clarifying the requirements of the phishing email detection system. This period will also involve organising access to the necessary tools, datasets, and development environment.

Following this, a research phase will take place in which literature related to phishing techniques, email-based attacks, and detection methodologies will be reviewed. This stage will be conducted gradually over the early part of the project to ensure a strong theoretical foundation. During the same period, relevant datasets both phishing and legitimate emails will be identified, acquired, and prepared for analysis.

As the project progresses, work will shift toward feature analysis and initial system design. During this stage, methods for extracting indicators such as URLs, headers, text cues, and sender identity patterns will be explored. Parallel to this, preliminary experiments with classification approaches (e.g., machine learning or rule-based detection) will be conducted to evaluate their suitability.

The development phase will begin once the system architecture is outlined. Throughout the middle portion of the project timeline, the phishing detection tool will be implemented, integrating feature extraction, analysis logic, and classification components. This work will be iterative, with adjustments based on ongoing discussions with the supervisor.

Testing and evaluation activities will be carried out in the later stages. The system's performance will be assessed using appropriate metrics, and refinements will be made where necessary. This phase will also include preparing documentation describing the methodology, results, limitations, and recommendations.

Toward the final stage, the completed system, report, and supporting materials will be refined and prepared for submission and presentation, following the general expectations of the project schedule.

17. Student Signature:

A handwritten signature in black ink, appearing to read "Shamsudin", written over a circular stamp or seal.

18. Supervisor Signature: [M. Dunja](#)

19. Date: 05-12-2025